# A reduction theorem for highest weight modules over toroidal Lie algebras

Ivan Dimitrov, Vyacheslav Futorny and Ivan Penkov

**Abstract**

Let $\mathfrak{g} = \overset{\circ}{\mathfrak{g}}[t_1^{\pm 1}, \dots, t_n^{\pm 1}] \oplus \mathcal{D} \oplus \mathfrak{z}$ be a toroidal Lie algebra corresponding to a semisimple Lie algebra $\overset{\circ}{\mathfrak{g}}$. We describe all Borel subalgebras of $\mathfrak{g}$ which contain the Cartan subalgebra $\mathfrak{h} = \overset{\circ}{\mathfrak{h}} \oplus \mathcal{D} \oplus \mathfrak{z}$, where $\overset{\circ}{\mathfrak{h}}$ is a fixed Cartan subalgebra of $\overset{\circ}{\mathfrak{g}}$. We show that each such Borel subalgebra $\mathfrak{b}$ determines a parabolic decomposition $\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{k} \oplus \mathfrak{n}^+$, where $\mathfrak{k}$ is a proper toroidal subalgebra of $\mathfrak{g}$ and $\mathfrak{b} \subset \mathfrak{k} \oplus \mathfrak{n}^+$. Our first main result is that, for any weight $\lambda$ which does not vanish on $\mathfrak{z} \cap [\mathfrak{k}, \mathfrak{k}]$, an arbitrary subquotient of the Verma module $M_{\mathfrak{g}}(\lambda)$ is induced from its $\mathfrak{k} \oplus \mathfrak{n}^+$–submodule of $\mathfrak{n}^+$–invariant vectors. This reduces the study of subquotients of $M_{\mathfrak{g}}(\lambda)$ to the study of subquotients of Verma modules over $\mathfrak{k}$. We then introduce categories $\mathcal{O}_{\mathfrak{k}}$ and $\mathcal{O}_{\mathfrak{g}}$ and their respective blocks $\mathcal{O}_{\mathfrak{g}}(\nu)$ and $\mathcal{O}_{\mathfrak{k}}(\nu)$ corresponding to a central charge $\nu$ which is nonzero on $\mathfrak{z} \cap [\mathfrak{k}, \mathfrak{k}]$. Our second main result is that the functors of induction and $\mathfrak{n}^+$–invariants are mutually inverse equivalences of the category $\mathcal{O}_{\mathfrak{k}}(\nu)$ and the full subcategory of $\mathcal{O}_{\mathfrak{g}}(\nu)$ whose objects are generated by their $\mathfrak{n}^+$–invariants.

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# Introduction

Toroidal Lie algebras are universal central extensions of iterated loop algebras with $n > 1$ variables and are a most natural generalization of affine Lie algebras. They were introduced by R. Moody, S. Rao and T. Yokonuma in [MRY] and since then have attracted considerable mathematical attention. Toroidal Lie algebras have also appeared in mathematical physics. T. Inami, H. Kanno, T. Ueno and C.-S. Xiong, [IKUX], have shown that a toroidal Lie algebra with $n = 2$ arises as a symmetry algebra in a four dimensional nonlinear sigma model with a Wess–Zumino like term introduced by S. Donaldson, [D], much in the same way as affine Lie algebras arise as symmetry algebras in the Wess–Zumino–Witten model. T. Inami, H. Kanno and T. Ueno, [IKU], have generalized this construction to arbitrary $n$.

A natural construction of representations of a toroidal Lie algebra is via induction from parabolic subalgebras whose Levi components are proper toroidal subalgebras. This construction has been carried out in special cases and the resulting modules have been studied

by S. Berman, Y. Billig, B. Cox, V. Futorny and I. Kashuba, see [BB], [BC], [CF], [FK]. Furthermore, in several of these works a natural phenomenon of reduction to modules over a proper toroidal subalgebra has been singled out. It generalizes the reduction to an affine or Heisenberg subalgebra known to exist for highest weight modules for nonstandard Borel subalgebras of affine Lie algebras, see [C] and [F1].

The purpose of the present paper is to study arbitrary highest weight modules over toroidal Lie algebras and to understand the reduction phenomenon in general. No general definition of a Borel subalgebra has appeared in the literature on toroidal algebras before. Our first observation is that since a standard Cartan subalgebra $\mathfrak{h}$ acts semisimply on a toroidal Lie algebra $\mathfrak{g}$ (i.e. $\mathfrak{g}$ has a root decomposition with respect to $\mathfrak{h}$), the definition of a Borel subalgebra of a Lie algebra with a root decomposition given in [DP1] and [DP2] applies to $\mathfrak{g}$. We give a complete combinatorial description of all Borel subalgebra of $\mathfrak{g}$ containing a fixed $\mathfrak{h}$. In particular, for $n > 1$, there are infinitely many nonconjugate Borel subalgebras (this was already noticed in [CF]). Furthermore, using the combinatorics of the root system of $\mathfrak{g}$, we assign to each Borel subalgebra a proper toroidal subalgebra $\mathfrak{k} \subset \mathfrak{g}$. (As $\mathfrak{k}$ can be finite dimensional, we allow $n$ to equal 0 in the definition of a toroidal Lie algebra.) In this way we obtain a parabolic decomposition $\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{k} \oplus \mathfrak{n}^+$ with $\mathfrak{b} \subset \mathfrak{k} \oplus \mathfrak{n}^+$. While $\mathfrak{k} = \mathfrak{h}$ for a generic $\mathfrak{b}$, for special choices of $\mathfrak{b}$ the subalgebra $\mathfrak{k}$ may equal almost any toroidal subalgebra of $\mathfrak{g}$ (for a more detailed discussion see Section 2).

The first main result of the present paper is that, if $\mathfrak{k}$ is infinite dimensional and the restriction of $\lambda$ onto the center $\mathfrak{z}$ of $\mathfrak{g}$ intersected with $[\mathfrak{k}, \mathfrak{k}]$ is nonzero, then any subquotient $M$ of the Verma module $M_{\mathfrak{g}}(\lambda)$ is isomorphic to the $\mathfrak{g}$–module induced from the $\mathfrak{k} \oplus \mathfrak{n}^+$–submodule $M^{\mathfrak{n}^+} \subset M$ consisting of all $\mathfrak{n}^+$–invariant vectors in $M$. This reduces the study of $M$ to the study of the $\mathfrak{k}$–module $M^{\mathfrak{n}^+}$, and is a vast generalization of previously obtained results in [BC], [CF] and [FK]. Our theorem implies that, if a Borel subalgebra $\mathfrak{b}$ yields an infinite dimensional toroidal subalgebra $\mathfrak{k}$, then only very special $\mathfrak{b}$–highest weight modules or their submodules are not parabolically induced from $\mathfrak{k}$: for those the highest weight $\lambda$ must vanish on the part of the center of $\mathfrak{g}$ contained in $[\mathfrak{k}, \mathfrak{k}]$. In fact, explicit vertex operator constructions of such modules have been given by S. Berman and Y. Billig, [BB], and it has been shown that they can be integrable with finite dimensional weight spaces. Moreover, we may speculate that the generic Borel subalgebras, i.e. those for which $\mathfrak{k}$ is finite dimensional and hence our result does not apply, yield mostly irreducible Verma modules. This may deserve further study, however it is beyond the scope of the present paper.

Our reduction result yields a natural equivalence of categories. We define categories $\mathcal{O}_{\mathfrak{g}}$ and $\mathcal{O}_{\mathfrak{k}}$ of $\mathfrak{g}$– and $\mathfrak{k}$–modules respectively, which admit an ascending filtration whose associated quotients are isomorphic to subquotients of Verma modules. We then consider their respective blocks $\mathcal{O}_{\mathfrak{g}}(\nu)$ and $\mathcal{O}_{\mathfrak{k}}(\nu)$ of modules with a central charge $\nu$. The second main result of the paper is that, for any $\nu$ which does not vanish on $\mathfrak{z} \cap [\mathfrak{k}, \mathfrak{k}]$, the functors of parabolic induction and the functor of $\mathfrak{n}^+$–invariants are mutually inverse equivalences of $\mathcal{O}_{\mathfrak{k}}(\nu)$ and the full subcategory of $\mathcal{O}_{\mathfrak{g}}(\nu)$ whose objects are generated by their $\mathfrak{n}^+$–invariants. Any irreducible object of $\mathcal{O}_{\mathfrak{g}}(\nu)$ is also an object of the latter subcategory, but as we show,

there are reducible modules in $\mathcal{O}_{\mathfrak{g}}(\nu)$ which are not generated by their $\mathfrak{n}^+$–invariants.

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**Notational Conventions**
The ground field is $\mathbb{C}$ and all Lie algebras and their representations are automatically assumed to be defined over $\mathbb{C}$. Some real vector spaces are also considered but this is clearly indicated. The superscript $^*$ stands for dual space, $U(\cdot)$ stands for enveloping algebra, $\mathrm{span}_A$ stands for span over a submonoid $A \subset \mathbb{C}$. We put $\mathbb{R}_+ := \{r \in \mathbb{R} \,|\, r \geq 0\}$ and $\mathbb{Z}_+ := \mathbb{Z} \cap \mathbb{R}_+$. Finally, a cone in a real vector space $W$ is by definition an $\mathbb{R}_+$–invariant submonoid $C \subset W$.

# 1 Triangular decompositions and highest weight modules

Let $W$ be a finite dimensional real vector space. For any subset $P \subset W\backslash\{0\}$, a decomposition $P = P^+ \sqcup P^-$ is called a triangular decomposition of $P$, if the cone $C^+ \subset W$ generated by $P^+ \cup (-P^-)$ (or, equivalently, its opposite cone $C^-$) contains no nontrivial vector subspace of $W$, cf. [DP2]. Let $F = \{\{0\} = F_n \subset F_{n-1} \subset \ldots \subset F_1 \subset F_0 = W\}$ be a flag of maximal length in $W$, where $n = \dim W$. We call $F$ oriented if, for each $i$, one of the connected components of $F_i\backslash F_{i+1}$ is labeled by $+$ and the other one is labeled by $-$, i.e. $F_i\backslash F_{i+1} = (F_i\backslash F_{i+1})^+ \sqcup (F_i\backslash F_{i+1})^-$. An oriented flag $F$ determines the triangular decomposition $P^\pm := P \cap (\cup_i (F_i\backslash F_{i+1})^\pm)$ of any subset $P \subset W\backslash\{0\}$. It is easy to see that if $P$ is finite, every triangular decomposition of $P$ can be determined by an oriented flag $F$ for which $P \cap F_1 = \emptyset$. This is no longer true if $P$ is infinite.

We now introduce an invariant of a triangular decomposition. For a subset $M \subset W$, let $\overline{M}$ denote the closure of $M$ in the natural topology of $W$. Given a triangular decomposition $P = P^+ \sqcup P^-$, set $P_0 := (\overline{C^+} \cap \overline{C^-}) \cap P$. The following statements are an exercise on closed convex sets.

**Proposition 1** *Let $P = P^+ \sqcup P^-$ be a triangular decomposition of $P \subset W\backslash\{0\}$.*

(i) *For any oriented flag of maximal length $F = \{\{0\} = F_n \subset F_{n-1} \subset \ldots \subset F_1 \subset F_0 = W\}$ which determines the decomposition $P = P^+ \sqcup P^-$, we have $P_0 \subset F_1$.*

(ii) *There exists an oriented flag of maximal length $F = \{\{0\} = F_n \subset F_{n-1} \subset \ldots \subset F_1 \subset F_0 = W\}$ which determines the decomposition $P = P^+ \sqcup P^-$ and such that $P_0 = F_1 \cap P$.* $\square$

Let $\mathfrak{g}$ be a (complex) Lie algebra with a fixed finite dimensional abelian selfnormalizing

subalgebra $\mathfrak{h} \subset \mathfrak{g}$ for which $\mathfrak{g}$ decomposes as

$$\mathfrak{g} = \mathfrak{h} \oplus (\oplus_{\alpha \in \mathfrak{h}^* \setminus \{0\}} \mathfrak{g}^\alpha), \tag{1}$$

where $\mathfrak{g}^\alpha := \{g \in \mathfrak{g} \,|\, [h, g] = \alpha(h)g \text{ for every } h \in \mathfrak{h}\}$. The set $\Delta := \{\alpha \in \mathfrak{h}^* \setminus \{0\} \,|\, \mathfrak{g}^\alpha \neq 0\}$ is the root system of $\mathfrak{g}$. A Borel subalgebra $\mathfrak{b}$ of $\mathfrak{g}$ containing $\mathfrak{h}$ is by definition any subalgebra of the form $\mathfrak{b} = \mathfrak{h} \oplus (\oplus_{\alpha \in \Delta^+} \mathfrak{g}^\alpha)$ for some triangular decomposition $\Delta = \Delta^+ \sqcup \Delta^-$ of $\Delta$ in the real vector space $\mathrm{span}_{\mathbb{R}} \Delta$. The triangular decomposition $\Delta = \Delta^+ \sqcup \Delta^-$ corresponding to $\mathfrak{b}$ defines also the subalgebra $\mathfrak{g}_\mathfrak{b} := \mathfrak{h} \oplus (\oplus_{\alpha \in \Delta_0} \mathfrak{g}^\alpha)$ of $\mathfrak{g}$.

A $\mathfrak{g}$-module $V$ is a weight module (with respect to $\mathfrak{h}$) if $V = \oplus_{\mu \in \mathfrak{h}^*} V^\mu$ where $V^\mu := \{v \in V \,|\, h \cdot v = \mu(h)v \text{ for all } h \in \mathfrak{h}\}$. The support of $V$ is the set $\mathrm{supp}\, V := \{\mu \in \mathfrak{h}^* \,|\, V^\mu \neq 0\}$. Any submodule or quotient of a weight module is a weight module. The enveloping algebra $U(\mathfrak{g})$ is a weight module under the adjoint action of $\mathfrak{g}$ with $\mathrm{supp}\, U(\mathfrak{g}) = Q := \mathrm{span}_{\mathbb{Z}} \Delta$.

For $\lambda \in \mathfrak{h}^*$ denote by $v^\lambda$ the one dimensional $\mathfrak{b}$–module on which $\mathfrak{h}$ acts via $\lambda$ and $\mathfrak{b}^+$ acts by zero. The $\mathfrak{g}$–module $M_\mathfrak{g}(\lambda) := U(\mathfrak{g}) \otimes_{U(\mathfrak{b})} v^\lambda$ is the Verma module with $\mathfrak{b}$–highest weight $\lambda$. Note that $M_\mathfrak{g}(\lambda)$ is a weight module which is isomorphic to $U(\mathfrak{b}^-)$ as a $\mathfrak{b}^-$–module. Furthermore, $M_\mathfrak{g}(\lambda)$ admits a unique maximal proper $\mathfrak{g}$–submodule, and hence, a unique irreducible quotient module $L_\mathfrak{g}(\lambda)$. A highest weight module with $\mathfrak{b}$–highest weight $\lambda$ is defined as an arbitrary nonzero quotient of $M_\mathfrak{g}(\lambda)$.

# 2 Toroidal Lie algebras

Let $\Gamma$ be a finitely generated free abelian group (written additively), and let $\mathcal{A}(\Gamma)$ denote the group algebra of $\Gamma$, i.e. $\mathcal{A}(\Gamma) = \{\sum_{\gamma \in \Gamma} c_\gamma t^\gamma \,|\, c_\gamma \neq 0 \text{ for finitely many } \gamma\}$, where $t$ is a formal variable. Set $\mathcal{D} := \mathrm{Hom}(\Gamma, \mathbb{C})$. Given a finite dimensional reductive (possibly abelian) Lie algebra $\mathring{\mathfrak{g}}$, the following commutation relations endow $\mathring{\mathfrak{g}} \otimes \mathcal{A}(\Gamma) \oplus \mathcal{D}$ with a Lie algebra structure:

$$\begin{array}{rcl} [g \otimes t^\gamma, g' \otimes t^{\gamma'}] & = & [g, g'] \otimes t^{\gamma+\gamma'}, \\ [d, g \otimes t^\gamma] & = & d(\gamma) g \otimes t^\gamma, \\ [d, d'] & = & 0, \end{array}$$

where $g, g' \in \mathring{\mathfrak{g}}$, $\gamma, \gamma' \in \Gamma$, and $d, d' \in \mathcal{D}$. Let $\varphi$ be a nondegenerate symmetric invariant bilinear form on $\mathring{\mathfrak{g}}$, and let $\mathfrak{z} := \mathbb{C} \otimes_{\mathbb{Z}} \Gamma$. The toroidal Lie algebra $\mathfrak{t}(\mathring{\mathfrak{g}}, \Gamma, \varphi) := \mathring{\mathfrak{g}} \otimes \mathcal{A}(\Gamma) \oplus \mathcal{D} \oplus \mathfrak{z}$ is defined as the central extension of $\mathring{\mathfrak{g}} \otimes \mathcal{A}(\Gamma) \oplus \mathcal{D}$ via the commutation relation

$$[g \otimes t^\gamma + d, g' \otimes t^{\gamma'} + d'] = [g, g'] \otimes t^{\gamma+\gamma'} + d(\gamma')g' \otimes t^{\gamma'} - d'(\gamma) g \otimes t^\gamma + \delta_{\gamma, -\gamma'} \varphi(g, g')\, 1 \otimes \gamma.$$

**Example 1.** Let $\mathring{\mathfrak{g}}$ be simple and $\Gamma = \mathbb{Z}^n$. Then we can identify $\mathcal{A}(\Gamma)$ with $\mathbb{C}[t_1^{\pm 1}, \ldots, t_n^{\pm 1}]$ so that $t^{(k_1, \ldots, k_n)} = t_1^{k_1} \ldots t_n^{k_n}$ for $(k_1, \ldots, k_n) \in \mathbb{Z}^n$, and $\mathcal{D} \cong \mathrm{span}\{t_1 \frac{\partial}{\partial t_1}|_{t_1 = \ldots = t_n = 1}, \ldots, t_n \frac{\partial}{\partial t_n}|_{t_1 = \ldots = t_n = 1}\}$. Furthermore, let $\kappa$ be the Killing form of $\mathring{\mathfrak{g}}$. The toroidal algebra $\mathring{\mathfrak{g}}[n] := \mathfrak{t}(\mathring{\mathfrak{g}}, \mathbb{Z}^n, \kappa)$ is the

"standard" toroidal Lie algebra, cf. [MRY]. Finally, let $\mathring{\mathfrak{h}}$ be a Cartan subalgebra of $\mathring{\mathfrak{g}}$, $\varphi = \kappa_{|\mathring{\mathfrak{h}}\times\mathring{\mathfrak{h}}}$ and $\Gamma = \mathbb{Z}^n$. In this case $\mathfrak{t}(\mathring{\mathfrak{h}}, \mathbb{Z}^n, \varphi)$ is a Heisenberg subalgebra of $\mathring{\mathfrak{g}}[n]$.

**Proposition 2**

(i) *If $\mathring{\mathfrak{g}}$ is simple, then $\mathring{\mathfrak{g}}[n]$ is isomorphic to the universal central extension of $\mathring{\mathfrak{g}} \otimes \mathbb{C}[t_1^{\pm}, \dots, t_n^{\pm}] \oplus \mathcal{D}$.*

(ii) *If $\mathring{\mathfrak{g}} = (\oplus_{i=1}^m \mathring{\mathfrak{g}}_i) \oplus \mathring{\mathfrak{z}}$ is reductive with simple components $\mathring{\mathfrak{g}}_i$ and center $\mathring{\mathfrak{z}}$, then*

$$\mathfrak{t}(\mathring{\mathfrak{g}}, \Gamma, \varphi) \cong ((\oplus_{i=1}^m \mathfrak{t}(\mathring{\mathfrak{g}}_i, \Gamma_i, \varphi_{|\mathring{\mathfrak{g}}_i\times\mathring{\mathfrak{g}}_i})) \oplus \mathfrak{t}(\mathring{\mathfrak{z}}, \Gamma_0, \varphi_{|\mathring{\mathfrak{z}}\times\mathring{\mathfrak{z}}}))/J,$$

*where $\Gamma_i$ are copies of $\Gamma$, and $J$ is the central ideal generated by all differences of corresponding elements in $\Gamma_i$ and $\Gamma_j$ for $1 \le i \ne j \le m$.*

**Sketch of the proof.** Statement (i) follows from the fact that any universal central extension of the $\mathbb{Z}^n$-graded Lie algebra $\mathring{\mathfrak{g}} \otimes \mathbb{C}[t_1^{\pm 1}, \dots, t_n^{\pm 1}]$ is isomorphic to $\mathring{\mathfrak{g}} \otimes \mathbb{C}[t_1^{\pm 1}, \dots, t_n^{\pm 1}] \oplus \Omega_{\mathbb{C}[t_1^{\pm 1}, \dots, t_n^{\pm 1}]}/d\mathbb{C}[t_1^{\pm 1}, \dots, t_n^{\pm 1}]$ (see [K]). Statement (ii) is a simple exercise. □

The term toroidal Lie algebra is often used for a consequtive extension of $\mathring{\mathfrak{g}} \otimes \mathbb{C}[t_1^{\pm 1}, \dots, t_n^{\pm 1}]$ by an infinite dimensional center, and then by a (finite or infinite dimensional) derivations algebra. In the present paper we consider the "minimal nontrivial" extensions of $\mathring{\mathfrak{g}} \otimes \mathbb{C}[t_1^{\pm 1}, \dots, t_n^{\pm 1}]$ defined above.

Unless explicitly stated otherwise, we will assume that $\mathring{\mathfrak{g}}$ is reductive. Fix $\mathfrak{g} = \mathfrak{t}(\mathring{\mathfrak{g}}, \Gamma, \varphi)$. If $\mathring{\mathfrak{h}}$ is a Cartan subalgebra of $\mathring{\mathfrak{g}}$, then $\mathfrak{h} := \mathring{\mathfrak{h}} \oplus \mathcal{D} \oplus \mathfrak{z}$ is a Cartan subalgebra of $\mathfrak{g}$ (here and below we identify $\mathring{\mathfrak{h}}$ with $\mathring{\mathfrak{h}} \otimes \mathbb{C} \subset \mathfrak{g}$), and $\mathfrak{g}$ admits a root decomposition (1). To describe the root system $\Delta$ of $\mathfrak{g}$, notice that $\Delta \subset \mathring{\mathfrak{h}}^* \oplus \mathcal{D}^*$ as $\mathfrak{z}$ is in the center of $\mathfrak{g}$. Since $\Gamma$ is canonically embedded into $\mathcal{D}^* = (\mathrm{Hom}(\Gamma, \mathbb{C}))^*$, we will denote the elements of $\Gamma$ and their images in $\mathcal{D}^*$ by the same letters. Then

$$\Delta = (\mathring{\Delta} + \Gamma) \cup (\Gamma \backslash \{0\}), \tag{2}$$

as $\mathfrak{g}^{\alpha+\gamma} = \mathring{\mathfrak{g}}^{\alpha} \otimes t^{\gamma}$ and $\mathfrak{g}^{\gamma} = \mathring{\mathfrak{h}} \otimes t^{\gamma}$ for $\alpha \in \mathring{\Delta}$, $\gamma \in \Gamma$. In what follows we will also identify $\Gamma$ with the subset $1 \otimes_{\mathbb{Z}} \Gamma$ of $\mathfrak{z}$. This enables us to evaluate on $\Gamma$ linear functions defined on $\mathfrak{z}$. With the conventions above $\gamma \in \Gamma$ may stand for three different things but the meaning of $\gamma$ will be clear from the context. Namely, when we write $\gamma$ alone or $x \otimes t^{\gamma}$ we understand $\gamma$ as an element of $\Gamma$, when we write $\alpha + \gamma$ we understand $\gamma$ as an element of $\mathcal{D}^*$, and when we write $\varphi(\gamma)$ we understand $\gamma$ as an element of $\mathfrak{z}$.

# 3 The decomposition $\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{k} \oplus \mathfrak{n}^+$

Let $\mathfrak{b}$ be a Borel subalgebra of $\mathfrak{g}$ corresponding to a triangular decomposition $\Delta = \Delta^+ \sqcup \Delta^-$. Set $\Gamma_0 := \Gamma \cap (\Delta_0 \cup \{0\})$ and note that $\Gamma_0$ is a subgroup of $\Gamma$. Indeed, according to Proposition 1(ii), $\Delta_0 = \Delta \cap F_1$ for a suitable hyperplane $F_1 \subset \mathrm{span}_{\mathbb{R}} \Delta$. Hence $\Gamma_0 = \Gamma \cap (\Delta_0 \cup \{0\}) = \Gamma \cap ((\Delta \cap F_1) \cup \{0\}) = ((\Gamma \cap F_1) \cap \Delta) \cup \{0\} = \Gamma \cap F_1$, i.e. $\Gamma_0$ equals the subgroup $\Gamma \cap F_1$ of $\Gamma$. For every $\alpha \in \mathring{\Delta}$, $(\alpha + \Gamma) \cap \Delta_0 \subset \alpha + \mathcal{D}^*$ is either empty or a translation of $\Gamma_0$. Denote by $\mathring{\Delta}_{\mathfrak{k}}$ the set of all roots $\alpha \in \mathring{\Delta}$ for which $(\alpha + \Gamma) \cap \Delta_0 \subset \alpha + \mathcal{D}^*$ is a translation of $\Gamma_0$, and let $\mathring{\mathfrak{k}} := \mathring{\mathfrak{h}} \oplus (\oplus_{\alpha \in \mathring{\Delta}_{\mathfrak{k}}} \mathring{\mathfrak{g}}^{\alpha})$.

**Proposition 3** *The Lie algebra $\mathring{\mathfrak{k}}$ is a reductive subalgebra of $\mathring{\mathfrak{g}}$. Furthermore, $\mathfrak{g}_{\mathfrak{b}} = \mathfrak{h} \oplus (\oplus_{\alpha \in \Delta_0} \mathfrak{g}^{\alpha})$ is a proper toroidal subalgebra of $\mathfrak{g}$ isomorphic to $\mathfrak{t}(\mathring{\mathfrak{k}}, \Gamma_0, \varphi_{|\mathring{\mathfrak{k}} \times \mathring{\mathfrak{k}}})$.*

**Proof.** We verify first that $\mathring{\mathfrak{k}}$ is a subalgebra of $\mathring{\mathfrak{g}}$. Indeed, if $\alpha, \beta \in \mathring{\Delta}_{\mathfrak{k}}$ and $\alpha + \beta \in \mathring{\Delta}$, then $\alpha + \gamma_1 \in F_1$ and $\beta + \gamma_2 \in F_1$, for some $\gamma_1, \gamma_2 \in \Gamma$, and hence $(\alpha + \beta) + (\gamma_1 + \gamma_2) \in F_1$, i.e. $\alpha + \beta \in \mathring{\Delta}_{\mathfrak{k}}$. Furthermore, $\mathring{\Delta}_{\mathfrak{k}} = -\mathring{\Delta}_{\mathfrak{k}}$, and therefore $\mathring{\mathfrak{k}} \subset \mathring{\mathfrak{g}}$ is reductive. The fact that $\varphi_{|\mathring{\mathfrak{k}} \times \mathring{\mathfrak{k}}}$ is nondegenerate follows from the fact that $\varphi_{|\mathring{\mathfrak{h}} \times \mathring{\mathfrak{h}}}$ and $\varphi_{|\mathring{\mathfrak{g}}^{\alpha} \times \mathring{\mathfrak{g}}^{-\alpha}}$, for $\alpha \in \mathring{\Delta}$, are nondegenerate. Note also that $\mathfrak{g}_{\mathfrak{b}}$ is a proper subalgebra of $\mathfrak{g}$ as $\Delta_0$ is a proper subset of $\Delta$.

To establish an isomorphism between $\mathfrak{g}_{\mathfrak{b}}$ and $\mathfrak{t}(\mathring{\mathfrak{k}}, \Gamma_0, \varphi_{|\mathring{\mathfrak{k}} \times \mathring{\mathfrak{k}}})$, fix a system of simple roots $\{\alpha_1, \ldots, \alpha_s\}$ of $\mathring{\Delta}_{\mathfrak{k}}$, and for every $1 \leq i \leq s$ fix a nonzero element $g^{\pm \alpha_i} \in \mathring{\mathfrak{g}}^{\pm \alpha_i}$. Furthermore, for every $1 \leq i \leq s$, fix $\gamma_i \in \Gamma$ such that $\alpha_i + \gamma_i \in \Delta_0$. Denote by $\mathring{\mathfrak{k}}'$ the subalgebra of $\mathfrak{g}_{\mathfrak{b}}$ generated by $g_i^{\pm \alpha_i} \otimes t^{\gamma_i}$ for $1 \leq i \leq s$ and $\mathfrak{h}$. Then there exists a unique Lie algebra isomorphism $\theta : \mathring{\mathfrak{k}} \to \mathring{\mathfrak{k}}'$ with $\theta(g^{\pm \alpha_i}) = g^{\pm \alpha_i} \otimes t^{\gamma_i}$ and such that $\theta_{|\mathring{\mathfrak{z}}_{\mathfrak{k}} \otimes \mathbb{C} \oplus \mathcal{D} \oplus \mathring{\mathfrak{z}}}$ is the identity, where $\mathring{\mathfrak{z}}_{\mathfrak{k}}$ denotes the center of $\mathring{\mathfrak{k}}$. The reader will check immediately that the extension $\Theta : \mathfrak{t}(\mathring{\mathfrak{k}}, \Gamma_0, \varphi_{|\mathring{\mathfrak{k}} \times \mathring{\mathfrak{k}}}) \to \mathfrak{g}_{\mathfrak{b}}$ of $\theta$ defined by $\Theta(g \otimes t^{\gamma}) := \theta(g) \otimes t^{\gamma}$ for $g \in \mathring{\mathfrak{k}}$ and $\gamma \in \Gamma_0$ is an isomorphism. □

When the Borel subalgebra $\mathfrak{b}$ of $\mathfrak{g}$ is given, we will denote $\mathfrak{g}_{\mathfrak{b}}$ by $\mathfrak{k}$, and the root system $\Delta_0$ of $\mathfrak{k}$ by $\Delta_{\mathfrak{k}}$. We have $\Delta_{\mathfrak{k}} = (\mathring{\Delta}_{\mathfrak{k}} + \Gamma_0) \cup (\Gamma_0 \backslash \{0\})$. In particular, $\Gamma_0 \neq 0$ precisely when $\mathfrak{k}$ is infinite dimensional. Set also $\mathfrak{n}^{\pm} = \oplus_{\alpha \in \Delta^{\pm} \backslash \Delta_{\mathfrak{k}}} \mathfrak{g}^{\alpha}$. Then

$$\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{k} \oplus \mathfrak{n}^+.$$

It is a natural problem to determine all pairs $(\Gamma_0, \mathring{\mathfrak{k}})$ arising from Borel subalgebras $\mathfrak{b}$ of $\mathfrak{g}$ as in Proposition 3 above. We will now reduce this problem to a purely combinatorial question about root systems. Although we do not answer the latter, we show that the

set of such pairs is rather large. An example of a pair which does not arise from a Borel subalgebra of $\mathfrak{g}$ is given below, see Example 3. Assume $\mathfrak{b}$ is determined by the partition $\Delta = \Delta^+ \sqcup \Delta^-$. Fix a hyperplane $F_1 \subset \operatorname{span}_{\mathbb{R}} \Delta$ corresponding to $\Delta = \Delta^+ \sqcup \Delta^-$ as in Proposition 1(ii). Let $F_1 = \ker \tau$, where $\tau : \operatorname{span}_{\mathbb{R}} \Delta \to \mathbb{R}$ is a nonzero linear function. Notice that $\operatorname{span}_{\mathbb{R}} \Delta = (\operatorname{span}_{\mathbb{R}} \Gamma) \oplus (\operatorname{span}_{\mathbb{R}} \mathring{\Delta})$, and hence $\tau = \tau_1 + \tau_2$, where $\tau_1 : \operatorname{span}_{\mathbb{R}} \Gamma \to \mathbb{R}$ and $\tau_2 : \operatorname{span}_{\mathbb{R}} \mathring{\Delta} \to \mathbb{R}$ are the restrictions of $\tau$ to $\operatorname{span}_{\mathbb{R}} \Gamma$ and $\operatorname{span}_{\mathbb{R}} \mathring{\Delta}$ respectively. Then $\Gamma_0 = \ker \tau_1 \cap \Gamma$ and $\mathring{\Delta}_{\mathfrak{k}} = \tau_2^{-1}(\tau_1(\Gamma)) \cap \mathring{\Delta}$. Moreover, $\tau_1(\Gamma)$ is a discrete subgroup of $\mathbb{R}$ of rank $d := \operatorname{rk} \Gamma - \operatorname{rk} \Gamma_0$. There are three possible cases.

(i) $\Gamma_0 = \Gamma$. Then $\mathring{\Delta}_{\mathfrak{k}} = \tau_2^{-1}(\{0\})$ for a nonzero $\tau_2$, i.e. the root system of $\mathring{\mathfrak{k}}$ is the intersection of $\mathring{\Delta}$ with a proper subspace of $\operatorname{span}_{\mathbb{R}} \mathring{\Delta}$.

(ii) $\operatorname{rk} \Gamma = 1$ and $\operatorname{rk} \Gamma_0 = 0$. Then $\mathring{\Delta}_{\mathfrak{k}} = \emptyset$.

(iii) None of the above holds for $\Gamma$ and $\Gamma_0$. In this case finding all possible pairs $(\Gamma_0, \mathring{\mathfrak{k}})$ is equivalent to finding all root subsystems of $\mathring{\Delta}$ of the form $\tau_2^{-1}(\Gamma') \cap \mathring{\Delta}$, where $\Gamma'$ is a subgroup of $\mathbb{R}$ of rank $d$.

**Example 3.**

1. Let $\Phi$ be a basis of $\mathring{\Delta}$, $\Phi_0 \subset \Phi$ and let $\mathring{\Delta}_{\mathfrak{k}}$ be the root subsystem of $\mathring{\Delta}$ with basis $\Phi_0$. Then $\mathring{\Delta}_{\mathfrak{k}} = \ker \tau_2 \cap \mathring{\Delta}$ for an appropriate $\tau_2$. Hence the pair $(\Gamma_0, \mathring{\mathfrak{k}})$ arises from a Borel sublagebra of $\mathfrak{g}$ unles $\Gamma_0 = \Gamma$ and $\Phi_0 = \Phi$, or $\operatorname{rk} \Gamma = 1$, $\operatorname{rk} \Gamma_0 = 0$ and $\Phi_0 = \emptyset$.

2. Let $\operatorname{rk} \Gamma_0 = \operatorname{rk} \Gamma - 1 \neq 0$. Then $\tau_1(\Gamma) \cong \mathbb{Z}$ and the condition on $\mathring{\Delta}_{\mathfrak{k}}$ is $\mathring{\Delta}_{\mathfrak{k}} = \tau_2^{-1}(\mathbb{Z}) \cap \mathring{\Delta}$ for some $\tau_2$. In particular, if $\mathring{\mathfrak{g}}$ is simple of type $B_2$ with $\mathring{\Delta} = \{\pm\varepsilon_1 \pm \varepsilon_2, \pm\varepsilon_1, \pm\varepsilon_2\}$, then $\tau_2^{-1}(\tau_1(\Gamma)) \cap \mathring{\Delta}$ can be any proper subsystem of $\mathring{\Delta}$. Thus $\mathring{\mathfrak{k}} = sl(2) \oplus sl(2)$ (when $\mathring{\Delta}_{\mathfrak{k}} = \{\pm\varepsilon_1 \pm \varepsilon_2\}$), $\mathring{\mathfrak{k}} \cong sl(2)$ or $\mathring{\mathfrak{k}} = \mathring{\mathfrak{h}}$. However, if $\mathring{\mathfrak{g}}$ is simple of type $C_3$, the reader will check that the subsystem $\mathring{\Delta}'$ consisting of the long roots cannot be realized as $\mathring{\Delta}_{\mathfrak{k}}$. Furthermore, it is not difficult to verify that for any subalgebra $\mathring{\mathfrak{k}}$ of $\mathring{\mathfrak{g}}$ the pair $(\Gamma_0, \mathring{\mathfrak{k}})$ arises from a Borel subalgebra of $\mathfrak{g}$ whenever $\operatorname{rk} \Gamma - \operatorname{rk} \Gamma_0$ is large enough.

Finally, note that the construction of $\mathfrak{k}$ admits natural iteration. Indeed, the Borel subalgebra $\mathfrak{b}$ determines a triangular decomposition $\Delta_0 = \Delta_0^+ \sqcup \Delta_0^-$, i.e. a Borel subalgebra of $\mathfrak{k}$. By applying the above construction to $\mathfrak{k}$ and its Borel subalgebra $\mathfrak{k} \cap \mathfrak{b}$, we obtain a toroidal subalgebra $\mathfrak{k}_2$ of $\mathfrak{k}$. Continuing the process, we obtain a strictly descending sequence of toroidal Lie algebras

$$\mathfrak{h} = \mathfrak{k}_s \subset \ldots \subset \mathfrak{k}_1 = \mathfrak{k} \subset \mathfrak{g} \tag{3}$$

together with a fixed Borel subalgebra $\mathfrak{k}_i \cap \mathfrak{b}$ of $\mathfrak{k}_i$ for each $1 \leq i \leq s$.

**Example 4.** Here is an example of two Borel subalgebras in $\mathfrak{g} = sl(3)[3]$ for which $\mathfrak{k}$ is the same but the corresponding sequences (3) differ. Let $\overset{\circ}{\Delta} = \{\pm\alpha, \pm\beta, \pm(\alpha+\beta)\}$ and let $\delta_1, \delta_2, \delta_3$ be fixed generators of $\Gamma$. Define the linear function $\tau : \mathrm{span}_{\mathbb{R}}\,\Delta \to \mathbb{R}$ by putting $\tau(\alpha) = 0$, $\tau(\beta) = \sqrt{2}$, $\tau(\delta_1) = 0$, $\tau(\delta_2) = 1$, $\tau(\delta_3) = \sqrt{3}$. It determines a partition $\Delta = \Delta^{--} \sqcup \Delta' \sqcup \Delta^{++}$, where $\Delta' := \ker\tau \cap \Delta$ and $\Delta^{++} := \{\xi \in \Delta \mid \tau(\xi) > 0\}$, $\Delta^{--} := -\Delta^{++}$. Clearly, $\Delta' = (\mathbb{Z}\delta_1 \backslash \{0\}) \cup (\pm\alpha + \mathbb{Z}\delta_1)$. Consider the following two triangular decompositions of $\Delta'$: $\Delta' := (\Delta')_1^+ \sqcup (\Delta')_1^-$ and $\Delta' := (\Delta')_2^+ \sqcup (\Delta')_2^-$, where $(\Delta')_1^+ := \{n\delta_1, \pm\alpha + n\delta_1 \mid n > 0\} \cup \{\alpha\}$, $(\Delta')_2^+ := \{n\delta_1 \mid n > 0\} \cup (\alpha + \mathbb{Z}\delta_1)$, $(\Delta')_i^- := -(\Delta')_i^+$. Finally, set $(\Delta)_i^\pm := \Delta^{\pm\pm} \sqcup (\Delta')_i^\pm$. One checks immediately that $\Delta = (\Delta)_i^+ \sqcup (\Delta)_i^-$ for $i = 1, 2$ are two triangular decompositions of $\Delta$. In both cases $\Delta_0 = \Delta'$ and $\mathfrak{k} \cong sl(2)[1]$ is an affine Lie algebra with roots $\Delta'$. The two triangular decompositions of $\Delta'$ produce respectively a standard and an imaginary Borel subalgebra of $\mathfrak{k}$. More precisely, the sequences corresponding to the decompositions $\Delta = (\Delta)_1^+ \sqcup (\Delta)_2^-$ and $\Delta = (\Delta)_2^+ \sqcup (\Delta)_2^-$ are respectively

$$\mathfrak{h} \subset \mathfrak{k} \cong sl(2)[1] \subset \mathfrak{g} \tag{4}$$

and

$$\mathfrak{h} \subset \mathfrak{k}_2 \cong \mathfrak{t}(\overset{\circ}{\mathfrak{h}}, \mathbb{Z}\delta_1, \kappa_{|\overset{\circ}{\mathfrak{h}}\times\overset{\circ}{\mathfrak{h}}}) \subset \mathfrak{k} \cong sl(2)[1] \subset \mathfrak{g}. \tag{5}$$

# 4 On the structure of subquotients of $M_{\mathfrak{g}}(\lambda)$

For the rest of the paper $\mathfrak{b}$ is a fixed Borel subalgebra of $\mathfrak{g} = \mathfrak{t}(\overset{\circ}{\mathfrak{g}}, \Gamma, \varphi)$ corresponding to the triangular decomposition $\Delta = \Delta^+ \sqcup \Delta^-$. We have $\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{k} \oplus \mathfrak{n}^+$, where $\mathfrak{n}^\pm := \oplus_{\alpha \in (\Delta^\pm \setminus \Delta_{\mathfrak{k}})} \mathfrak{g}^\alpha$. For $\lambda \in \mathfrak{h}^*$, set $D_{\mathfrak{k}}(\lambda) := \lambda + \mathrm{span}_{\mathbb{Z}}\,\Delta_{\mathfrak{k}}$ and put $M_{\mathfrak{k}} := \oplus_{\mu \in D_{\mathfrak{k}}(\lambda)} M^\mu$ for any subquotient $M$ of $M_{\mathfrak{g}}(\lambda)$.

In this section we assume that $\Gamma_0 \neq 0$, i.e. that $\mathfrak{k}$ is infinite dimensional, and establish a relation between submodules of $M_{\mathfrak{g}}(\lambda)$ and respective submodules of $M_{\mathfrak{k}}(\lambda)$. We consider first the case when $\overset{\circ}{\mathfrak{g}}$ is semisimple as here the result is very natural and easy to state.

**Theorem 1** *Let $\overset{\circ}{\mathfrak{g}}$ be semisimple, $\mathfrak{k}$ be infinite dimensional and $N$ be a nontrivial submodule of $M_{\mathfrak{g}}(\lambda)$. The following assertions hold.*

(i) $N_{\mathfrak{k}} \neq 0$.

(ii) *If $\lambda(\Gamma_0) \neq 0$, then the canonical $\mathfrak{g}$–module homomorphism $U(\mathfrak{g}) \otimes_{U(\mathfrak{k}\oplus\mathfrak{n}^+)} N_{\mathfrak{k}} \to N$ is an isomorphism.*

The following corollary is immediate.

**Corollary 1** *Let $\overset{\circ}{\mathfrak{g}}$ be semisimple and $\mathfrak{k}$ be infinite dimensional.*

(i) *For any $\lambda$ the Verma module $M_{\mathfrak{g}}(\lambda)$ is irreducible if and only if the Verma module $M_{\mathfrak{k}}(\lambda)$ is irreducible.*

(ii) *If $\lambda(\Gamma_0) \neq 0$, then $L_{\mathfrak{g}}(\lambda) \cong U(\mathfrak{g}) \otimes_{U(\mathfrak{k} \oplus \mathfrak{n}^+)} L_{\mathfrak{k}}(\lambda)$.*

The proof of Theorem 1 uses induction on the Lie algebra $\mathfrak{g}$ which requires a similar statement for reductive $\overset{\circ}{\mathfrak{g}}$ as well. Therefore, in the rest of the section we will state and prove an analog of Theorem 1 for reductive $\overset{\circ}{\mathfrak{g}}$. Theorem 1 is a particular case of Theorem 2 below.

Let $\overset{\circ}{\mathfrak{g}} = \overset{\circ}{\mathfrak{g}}^{\mathrm{ss}} \oplus \overset{\circ}{\mathfrak{z}}$ be reductive Lie algebra with semisimple part $\overset{\circ}{\mathfrak{g}}^{\mathrm{ss}}$ and center $\overset{\circ}{\mathfrak{z}}$. For $\lambda \in \mathfrak{h}^*$, put $\nu := \lambda_{|\mathfrak{z}}$ and

$$Z^-(\nu) := \mathrm{span}\{z \otimes t^\gamma \in \mathfrak{n}^- \mid z \in \overset{\circ}{\mathfrak{z}} \text{ and } \lambda(\gamma) = 0\}.$$

Then $\mathfrak{n}^- = \widetilde{\mathfrak{n}^-} \oplus Z^-(\nu)$, where the splitting of $\mathfrak{t}(\overset{\circ}{\mathfrak{h}}, \Gamma, \varphi_{|\overset{\circ}{\mathfrak{h}} \times \overset{\circ}{\mathfrak{h}}}) \cap \mathfrak{n}^-$ is compatible with the splitting $\overset{\circ}{\mathfrak{h}} = \overset{\circ}{\mathfrak{h}}^{\mathrm{ss}} \oplus \overset{\circ}{\mathfrak{z}}$. Set $\tilde{\mathfrak{g}} := \widetilde{\mathfrak{n}^-} \oplus \mathfrak{k} \oplus \mathfrak{n}^+$. One can consider $U(\tilde{\mathfrak{g}})$ as a left $U(\mathfrak{g})$–module with trivial action of $Z^-(\nu)$, and define the $\mathfrak{g}$–module $\widetilde{M}_{\mathfrak{g}}(\lambda)$ as $U(\tilde{\mathfrak{g}}) \otimes_{U(\mathfrak{b})} v^\lambda$.

**Theorem 2** *Let $\overset{\circ}{\mathfrak{g}}$ be reductive, $\mathfrak{k}$ be infinite dimensional and $N$ be a nontrivial submodule of $\widetilde{M}_{\mathfrak{g}}(\lambda)$. The following assertions hold.*

(i) *$N_{\mathfrak{k}} \neq 0$.*

(ii) *If $\lambda(\Gamma_0) \neq 0$, then the canonical $\mathfrak{g}$–module homomorphism $U(\tilde{\mathfrak{g}}) \otimes_{U(\mathfrak{k} \oplus \mathfrak{n}^+)} N_{\mathfrak{k}} \to N$ is an isomorphism.*

The following corollary is immediate.

**Corollary 2** *Let $\overset{\circ}{\mathfrak{g}}$ be reductive and $\mathfrak{k}$ be infinite dimensional.*

(i) *For any $\lambda$ the module $\widetilde{M}_{\mathfrak{g}}(\lambda)$ is irreducible if and only if the Verma module $M_{\mathfrak{k}}(\lambda)$ is irreducible.*

(ii) *If $\lambda(\Gamma_0) \neq 0$, then $L_{\mathfrak{g}}(\lambda) \cong U(\tilde{\mathfrak{g}}) \otimes_{U(\mathfrak{k} \oplus \mathfrak{n}^+)} L_{\mathfrak{k}}(\lambda)$.*

**Proof of Theorem 2.** Fix an oriented flag $F$ of maximal length which determines the triangular decomposition $\Delta = \Delta^+ \sqcup \Delta^-$ as in Proposition 1 (ii). Let $F_1 = \ker \tau$, i.e. $\alpha \in \Delta$ is a root of $\mathfrak{k}$ if and only if $\tau(\alpha) = 0$, cf. Section 3. Put $\mathfrak{b}^\pm = \oplus_{\alpha \in \Delta^\pm} \mathfrak{g}^\alpha$, and let $1_\lambda \in v^\lambda$ be a fixed nonzero vector.

Consider a decomposition $\mathring{\mathfrak{g}} = \oplus_p \mathring{\mathfrak{g}}_p$, where each $\mathring{\mathfrak{g}}_p$ is either a simple Lie algebra or is the one dimensional abelian Lie algebra $\mathbb{C}z$ for some $z \in \mathring{\mathfrak{z}}$ with $\varphi(z,z) \neq 0$. As a first step we show that it suffices to establish the Theorem for each of the algebras $\mathfrak{g}_p := \mathfrak{t}(\mathring{\mathfrak{g}}_p, \Gamma, \varphi_{|\mathring{\mathfrak{g}}_p \times \mathring{\mathfrak{g}}_p})$ (cf. Proposition 2 (ii)). Set $\bar{\mathring{\mathfrak{g}}}_p := \oplus_{i \neq p} \mathring{\mathfrak{g}}_i$, $\bar{\mathfrak{g}}_p := \mathfrak{t}(\bar{\mathring{\mathfrak{g}}}_p, \Gamma, \varphi_{|\bar{\mathring{\mathfrak{g}}}_p \times \bar{\mathring{\mathfrak{g}}}_p})$ and assume that Theorem 2 is true for $\mathfrak{g}_1$ and, by induction, also for $\bar{\mathfrak{g}}_1$. Let $X \in N$ be a nonzero weight vector written as $X = (\sum_i u_i v_i w_i) \cdot 1_\lambda$, where $u_i \in U(\mathfrak{n}^- \cap \mathfrak{g}_1)$, $v_i \in U(\mathfrak{k} \cap \mathfrak{g}_1)$, $w_i \in U(\mathfrak{b}^- \cap \bar{\mathfrak{g}}_1)$ and the vectors $w_i \cdot 1_\lambda$ are linearly independent. Let $M_i$ denote the $\mathfrak{g}_1$–module generated by $w_i \cdot 1_\lambda$. Then $M_i \cong \widetilde{M}_{\mathfrak{g}_1}(\lambda_{|\mathfrak{h} \cap \mathfrak{g}_1})$. Set $X_i := (u_i v_i w_i) \cdot 1_\lambda$. The $\mathfrak{g}_1$–module $N_i$ generated by $X_i$ is a submodule of $M_i$. By Theorem 2 (i) applied to the $\mathfrak{g}_1$–modules $N_1$ and $M_1$, there exists $\bar{u} \in U(\mathfrak{g}_1)$ for which $\bar{u} \cdot X_1 = (\bar{u} u_1 v_1) \cdot (w_1 \cdot 1_\lambda)$ is a nonzero weight vector in $(N_1)_{\mathfrak{k} \cap \mathfrak{g}_1}$. But then $\bar{u} \cdot X \neq 0$ and, clearly, $\bar{u} \cdot X \in N_{\mathfrak{k} \cap \mathfrak{g}_1}$. Repeating the above argument for $\bar{\mathfrak{g}}_1$ instead of $\mathfrak{g}_1$ and $\bar{u} \cdot X$ instead of $X$, we prove that $N_{\mathfrak{k}} \neq 0$. If we furthermore assume that $\lambda(\Gamma_0) \neq 0$, Theorem 2 (ii) applied to the $\mathfrak{g}_1$–modules $N_i$ and $M_i$ implies that $v_i \cdot (w_i \cdot 1_\lambda) \in N_{\mathfrak{k} \cap \mathfrak{g}_1}$, and hence $X$ belongs to the $\mathfrak{g}_1$–module generated by $\oplus_i (N_i)_{\mathfrak{k} \cap \mathfrak{g}_1}$. A similar argument for $\bar{\mathfrak{g}}_1$ implies that $N$ is generated by $N_{\mathfrak{k}}$.

It now remains to prove the Theorem when $\mathring{\mathfrak{g}}$ is simple or one dimensional. First we consider the case when $\mathring{\mathfrak{g}} = \mathbb{C}z$ for some $z$ with $\varphi(z,z) \neq 0$. Let $X \in N$ be a nonzero vector. Write $X = (\sum_i u_i v_i) \cdot 1_\lambda$, where $u_i \in U(\mathfrak{n}^-)$ are disctinct monomials in $\{z \otimes t^{\gamma'} \mid \lambda(\gamma') \neq 0\}$ and $v_i \in U(\mathfrak{k})$. We will show that $v_i \in N_{\mathfrak{k}}$ for every $i$. If $z \otimes t^{\gamma'}$ occurs in some of the monomials $u_i$, then for any $r \in \mathbb{Z}_+$ we have

$$(z \otimes t^{-\gamma'})^r \cdot X = r!(-\varphi(z,z)\lambda(\gamma'))^r (\sum_i u_i' v_i) \cdot 1_\lambda,$$

where $u_i'$ is the monomial $u_i$ with $r$ terms equal to $z \otimes t^{\gamma'}$ erased whenever at least $r$ such terms occur in $u_i$, and $u_i' = 0$ otherwise. (In this case $U(\mathfrak{n}^-)$ is commutative so the procedure of erasing terms from monomials is not ambiguous.) Taking $r$ to be the maximal degree in which $z \otimes t^{\gamma'}$ occurs in a monomial $u_i$, we obtain the nonzero vector

$$X' := \frac{1}{r!(-\varphi(z,z)\lambda(\gamma'))^r} (z \otimes t^{-\gamma'})^r \cdot X = (\sum_j u_j' v_j') \cdot 1_\lambda \in N,$$

where $v_j'$ equals $v_i$ for some $i$ and $u_j'$ is obtained from $u_i$ by erasing all terms equal to $z \otimes t^{\gamma'}$. The same procedure applied to $X'$ yields the nonzero vector

$$X'' := \frac{1}{(r')!(-\varphi(z,z)\lambda(\gamma''))^{r'}} (z \otimes t^{-\gamma''})^{r'} \cdot X' = (\sum_k u_k'' v_k'') \cdot 1_\lambda \in N,$$

where $v_k''$ equals $v_i$ for some $i$ and $u_k''$ is obtained from $u_i$ by erasing all terms equal to $z \otimes t^{\gamma'}$ and $z \otimes t^{\gamma''}$. Continuing in this way, we obtain that $v_{i_0} \cdot 1_\lambda \in N$ for some $i_0$. Then we set $\bar{X} := X - (u_{i_0} v_{i_0}) \cdot 1_\lambda$ and show by an easy induction argument that $v_i \cdot 1_\lambda \in N$ for every $i$.

This completes the proof in the case when $\overset{\circ}{\mathfrak{g}}$ is abelian. Note that in this case the condition on $\lambda$ is not needed for assertion (ii).

For the rest of the proof $\overset{\circ}{\mathfrak{g}}$ is a simple Lie algebra and we fix an indivisible element $\gamma \in \Gamma_0$, which has the additional property $\lambda(\gamma) \neq 0$, whenever $\lambda(\Gamma_0) \neq 0$. (An element $\gamma' \in \Gamma_0$ is indivisible if $l\gamma' \in \Gamma$ for $l \in \mathbb{Q}$ implies $l \in \mathbb{Z}$.) The proof will be carried out in two separate cases: when $\operatorname{rk}\Gamma_0 \geq \operatorname{rk}\Gamma - 1$ and when $\operatorname{rk}\Gamma_0 < \operatorname{rk}\Gamma - 1$.

**Case 1:** $\operatorname{rk}\Gamma_0 \geq \operatorname{rk}\Gamma - 1$**.** Fix a basis $\overset{\circ}{B}$ of $\overset{\circ}{\mathfrak{g}}$ consisting of root vectors of $\overset{\circ}{\mathfrak{g}}$ and of a basis of $\overset{\circ}{\mathfrak{h}}$ with the property that for each $h \in \overset{\circ}{\mathfrak{h}} \cap \overset{\circ}{B}$ there are corresponding elements $e, f \in \overset{\circ}{B}$ so that $\{e, h, f\}$ forms a canonical $sl(2)$–basis. Then the set $B = \{x \otimes t^\gamma \in \mathfrak{b}^- \mid x \in \overset{\circ}{B}, \gamma \in \Gamma\}$ is a basis of $\mathfrak{b}^-$. For $x \in B$, let $\operatorname{rt}(x)$ denote the corresponding root.

Assume $N^\mu \neq 0$ and fix a nonzero vector $X \in N^\mu$. Write $X = (\sum_i u_i v_i) \cdot 1_\lambda$, where $u_i \in U(\mathfrak{n}^-)$ are disctinct monomials in the basis $B$ and $v_i \in U(\mathfrak{k})$. We are going to prove that

(i) $N_\mathfrak{k} \neq 0$;

(ii) if $\lambda(\Gamma_0) \neq 0$, then $v_i \in N_\mathfrak{k}$ for every $i$.

To prove (ii) it is enough to show that $v_{i_0} \cdot 1_\lambda \in N_\mathfrak{k}$ for some $i_0$ for which $u_{i_0}$ has maximal degree (here degree refers to the canonical filtration of $U(\mathfrak{n}^-)$) among all $u_i$. Indeed, then we will consider $X - u_{i_0} v_{i_0} \cdot 1_\lambda = (\sum_{i \neq i_0} u_i v_i) \cdot 1_\lambda$, for which there are fewer monomials $u_i$ whose degree equals the degree of $u_{i_0}$ and complete the argument by induction.

Fix a set of generators $\gamma_1, \ldots, \gamma_n$ of $\Gamma$ so that $\gamma_1, \ldots, \gamma_{n-1} \in \Gamma_0$. Recall that $Q = \operatorname{span}_\mathbb{Z} \Delta$ and notice that $\tau(Q) = \mathbb{Z}\tau(\gamma_n) + \tau(\overset{\circ}{\Delta})$ is a discrete subset of $\mathbb{R}$. Furthermore, fix a linear order on $\Delta^-$ with the property that $\beta \prec \alpha$ whenever one of the following conditions holds:

– for some $i$ $\alpha$ is a root of $\mathfrak{k}_i$ and $\beta$ is not a root of $\mathfrak{k}_i$;

– there is no such $i$, but $(\alpha, \gamma) < (\beta, \gamma)$.

Finally, fix a linear order on $B$ in a compatible way, i.e. $x \prec y$ whenever $\operatorname{rt}(x) \prec \operatorname{rt}(y)$.

The proof of (i) and (ii) will be carried out by induction on $\tau(\lambda - \mu)$. Denote by $T_X^-$ the set of roots $\alpha$ of $\mathfrak{n}^-$ for which there exists a root vector $x \in \mathfrak{g}^\alpha \cap B$ which occurs in a monomial $u_i$ of maximal degree, and let $T_X^0$ be the set of roots $\beta$ of $\mathfrak{k}$ for which there exists a root vector $y \in \mathfrak{g}^\beta \cap B$ which occurs in some $v_i$. Set $T_X := T_X^- \cup T_X^0$. Fix a root $\alpha_0 \in T_X^-$ that cannot be written as a nonnegative integer combination of roots from $(T_X \cup \{\gamma\}) \backslash \{\alpha_0\}$ (since $T_X$ is a finite set, such a root always exists). Now we consider two subcases.

**Subcase 1:** $\alpha_0 \notin \Gamma$**.** Then $\alpha_0 = \overset{\circ}{\alpha}_0 + \gamma_0$ for some $\overset{\circ}{\alpha}_0 \in \overset{\circ}{\Delta}$ and $\gamma_0 \in \Gamma$. Let $f \in \overset{\circ}{B}$ be the root vector corresponding to $\overset{\circ}{\alpha}_0$ and let $e \in \overset{\circ}{B}$ be the root vector corresponding to $-\overset{\circ}{\alpha}_0$. Set

$h := [e, f]$. We claim that the vector $X' := (e \otimes t^{-\gamma_0 + K\gamma}) \cdot X$ is nonzero for large enough $K \in \mathbb{Z}_+$. Indeed, assume that $f \otimes t^{\gamma_0}$ occurs in $u_1 = x_1 \dots x_p$, where $x_i \in B \cap \mathfrak{n}^-$ and $\alpha_0$ is among $\mathrm{rt}(x_1), \dots, \mathrm{rt}(x_p)$. Then

$$
\begin{aligned}
(e \otimes t^{-\gamma_0 + K\gamma}) \cdot (u_1 v_1 \cdot 1_\lambda) &= \\
(\textstyle\sum_{i=1}^{p} x_1 \dots x_{i-1} [e \otimes t^{-\gamma_0 + K\gamma}, x_i] x_{i+1} \dots x_p v_1) \cdot 1_\lambda &= \\
(x_1 \dots x_{j-1} x_{j+1} \dots x_p)((l\varphi(e,f))\,(h \otimes t^{K\gamma}) v_1) \cdot 1_\lambda + X'' &\ ,
\end{aligned} \tag{6}
$$

where $j$ is the minimal index for which $x_j = f \otimes t^{\gamma_0}$ and there are exactly $l$ such indices. Now write $X'$ as $X' = (\sum_j u_j' v_j') \cdot 1_\lambda$, where $u_1' := x_1 \dots x_{j-1} x_{j+1} \dots x_p$, see (6). We claim that $v_1' = (l\varphi(e,f))\,(h \otimes t^{K\gamma}) v_1$, i.e. that $u_1'$ can only be obtained from commuting vectors of the form $e \otimes t^{-\gamma_0 + K\gamma}$ with $f \otimes t^{\gamma_0}$ in (6). Indeed, this is a consequence of the choice of $\alpha_0$: $\mathrm{rt}(e \otimes t^{-\gamma_0 + K\gamma}) = -\alpha_0 + K\gamma$. On one hand, commuting $e \otimes t^{-\gamma_0 + K\gamma}$ consequtively with vectors among $x_1, \dots, x_p$ different from $f \otimes t^{\gamma_0}$ cannot result in a root vector whose corresponding root belongs to $\Gamma$. On the other hand, commuting $e \otimes t^{-\gamma_0 + K\gamma}$ consequtively with at least two vectors among $x_1, \dots, x_p$ of which at least one equals $f \otimes t^{\gamma_0}$ will result in a root vector of $\mathfrak{n}^-$ whose corresponding root $\beta$ has a product $(\beta, \gamma)$ larger than all such products corresponding to the vectors $x_i$. This proves that $X' \neq 0$, and clearly $X' \in N^{\mu'}$ with $\tau(\lambda - \mu') < \tau(\lambda - \mu)$. Hence, by induction, $N_{\mathfrak{k}} \neq 0$.

If, furthermore, $\lambda(\Gamma_0) \neq 0$, the induction assumption implies $1/(l\varphi(e,f))\, v_1' = ((h \otimes t^{K\gamma}) v_1) \cdot 1_\lambda \in N$. If $h \otimes t^{K\gamma} \notin \mathfrak{k}_2$, then $h \otimes t^{-K\gamma} \cdot (((h \otimes t^{K\gamma}) v_1) \cdot 1_\lambda) = -K\varphi(h,h)\lambda(\gamma) v_1 \cdot 1_\lambda \in N$, which shows that $v_1 \cdot 1_\lambda \in N$. The proof is complete in this subcase. If $h \otimes t^{K\gamma} \in \mathfrak{k}_2$, we use Theorem 2 for the proper subalgebra $\mathfrak{k}$ instead of $\mathfrak{g}$, its Borel subalgebra $\mathfrak{b} \cap \mathfrak{k}$, the corresponding Verma module $M_{\mathfrak{k}}(\lambda)$ and its submodule $N \cap M_{\mathfrak{k}}(\lambda)$. A computation similar to the one above shows that $v_1 \cdot 1_\lambda \in N$.

**Subcase 2:** $\alpha_0 = \gamma_0 \in \Gamma$**.** Fix a vector $h \otimes t^{\gamma_0}$ which occurs in a monomial $u_i$ of maximal degree and consider the vectors $e$ and $f$ of $\overset{\circ}{B}$ corresponding to $\mathfrak{h}$, see the convention about the choice of $\overset{\circ}{B}$. Denote the root of $\overset{\circ}{\mathfrak{g}}$ corresponding to $e$ by $\overset{\circ}{\alpha}$. Let $q$ be the smallest integer for which $\overset{\circ}{\alpha} + q\gamma_0$ is a root of $\mathfrak{n}^-$. Then, similarly to Subcase 1, consider $X' := (f \otimes t^{-q\gamma_0 + K\gamma}) \cdot X$ and write $X' = (\sum_j u_1' v_1') \cdot 1_\lambda$. There are two possibilities: $\alpha + (q-1)\gamma_0$ is a root either of $\mathfrak{n}^+$ or of $\mathfrak{k}$. An argument similar to the one used in Subcase 1 shows that, for large $K$,

- if $\alpha + (q-1)\gamma_0$ is a root of $\mathfrak{n}^+$, then we have (after possible relabeling) $u_1' = (f \otimes t^{-(q-1)\gamma_0 + K\gamma}) x_1 \dots x_{i-1} x_{i+1} \dots x_p$ and $v_1' = -2l\, v_1$, where $i$ is the minimal index for which $x_i = h \otimes t^{\gamma_0}$ and $l$ is the number of indices $r$ with $x_r = h \otimes t^{\gamma_0}$;
- if $\alpha + (q-1)\gamma_0$ is a root of $\mathfrak{k}$, then we have (after possible relabeling) $u_1' = x_1 \dots x_{i-1} x_{i+1} \dots x_p$ and $v_1' = (-2l)\,(f \otimes t^{-(q-1)\gamma_0 + K\gamma}) v_1$, where $i$ is the minimal index for which $x_i = h \otimes t^{\gamma_0}$ and $l$ is the number of indices $r$ with $x_r = h \otimes t^{\gamma_0}$.

The proof is completed by induction as in Subcase 1.

**Case 2:** $\operatorname{rk}\Gamma_0 < \operatorname{rk}\Gamma - 1$**.** In this case the set $\{\tau(\gamma) \mid \gamma \in \Gamma\}$ is a dense subset of $\mathbb{R}$. Fix a nonzero vector $X \in N$ and write $X = (\sum_i u_i v_i) \cdot 1_\lambda$, where $u_i \in U(\mathfrak{n}^-)$ are linearly independent and $v_i \in U(\mathfrak{k})$. Furthermore, fix $\overset{\circ}{\alpha} \in \overset{\circ}{\Delta}$ and a nonzero element $e \in \overset{\circ}{\mathfrak{g}}^{\alpha}$. We will prove first that $N$ contains nonzero vectors of the form $(\sum_i u_i' v_i') \cdot 1_\lambda$, where $u_i'$ are (distinct) monomials in $\{e \otimes t^{\gamma'} \in \mathfrak{n}^- \mid \gamma' \in \Gamma\}$ and all $v_i'$ are among $v_i$.

As in Case 1 we work with a basis $\overset{\circ}{B}$ of $\overset{\circ}{\mathfrak{g}}$ consisting of root vectors of $\overset{\circ}{\mathfrak{g}}$ and a basis of $\overset{\circ}{\mathfrak{h}}$. However, the basis for $\overset{\circ}{\mathfrak{h}}$ will be fixed later in the proof. Note that, since $\overset{\circ}{\mathfrak{h}}$ is abelian, $U(\mathfrak{t}(\overset{\circ}{\mathfrak{h}}, \Gamma, \varphi_{|\overset{\circ}{\mathfrak{h}} \times \overset{\circ}{\mathfrak{h}}}) \cap \mathfrak{n}^-)$ is simply the symmetric algebra of $\mathfrak{t}(\overset{\circ}{\mathfrak{h}}, \Gamma, \varphi_{|\overset{\circ}{\mathfrak{h}} \times \overset{\circ}{\mathfrak{h}}}) \cap \mathfrak{n}^-$, and hence all definitions below are independent of the choice of a basis of $\overset{\circ}{\mathfrak{h}}$. Consider a linear order on $\overset{\circ}{B}$ such that

- if $x_1$ and $x_2 \in \overset{\circ}{B}$ are root vectors with corresponding roots $\overset{\circ}{\beta}_1$ and $\overset{\circ}{\beta}_2$, then $(\overset{\circ}{\alpha}, \overset{\circ}{\beta}_1) < (\overset{\circ}{\alpha}, \overset{\circ}{\beta}_2)$ implies $x_1 \prec x_2$,
- if $x_1 \in \overset{\circ}{B}$ is a root vector with corresponding root $\overset{\circ}{\beta}_1$, and $x_2 \in \overset{\circ}{B} \cap \mathfrak{h}$, then $(\overset{\circ}{\alpha}, \overset{\circ}{\beta}_1) < 0$ implies $x_1 \prec x_2$ and $(\overset{\circ}{\alpha}, \overset{\circ}{\beta}_1) \geq 0$ implies $x_2 \prec x_1$.

We set $B := \{x \otimes t^{\gamma'} \in \mathfrak{b}^- \mid x \in \overset{\circ}{B}\}$ and extend the order on $\overset{\circ}{B}$ to a linear order on $B$, i.e. if $x_1 \prec x_2 \in \overset{\circ}{B}$, then $x_1 \otimes t^{\gamma'} \prec x_2 \otimes t^{\gamma''} \in B$. Any monomial $u = z_1 \dots z_l$ in $B$ is written so that if $z_i \prec z_j$, then $i < j$. The inverse lexicographical order on the monomials in $B$ has the following property: if $z_{l-i} \prec z'_{l'-i}$, where $i$ is the minimal nonnegative integer for which $z_{l-i}$ and $z'_{l'-i}$ correspond to different elements of $\overset{\circ}{B}$, then $u = z_1 \dots z_l \prec z'_1 \dots z'_{l'} = u'$. Also, if $u' = u''u$, with $u'' \neq 1$, then $u' \prec u$. Let $X = (\sum_i u_i v_i) \cdot 1_\lambda \in N^\mu$, where $u_i$ are distinct monomials in $B$. For $u = (x_1 \otimes t^{\gamma_1}) \dots (x_l \otimes t^{\gamma_l})$ we define the height of a monomial $u$ in $B$ as $\operatorname{ht}(u) := \sum_j (\overset{\circ}{\alpha} - \overset{\circ}{\beta}_j, \overset{\circ}{\alpha})$, where $x_j \in \overset{\circ}{\mathfrak{g}}^{\overset{\circ}{\beta}_j}$, assuming that $\overset{\circ}{\mathfrak{g}}^0 := \overset{\circ}{\mathfrak{h}}$. We are going to prove the existence of $(\sum_i u_i' v_i') \cdot 1_\lambda \in N$ as above by induction on the maximal height of the monomials $u_1, u_2, \dots$. If the maximal height is 0, there is nothing to prove. Otherwise, assume that the monomial $u_1$ has maximal height and is, in addition, maximal (with respect to the inverse lexicographical order) among all monomials of maximal height. Let $u_1 = x_1 \dots x_k \dots x_l$, where $k$ is the largest index for which the vector $z$ of $\overset{\circ}{B}$ corresponding to $x_k$ does not equal $e$. We now choose a root vector $e' \in \overset{\circ}{\mathfrak{g}}^{\overset{\circ}{\alpha}'}$ with $(\overset{\circ}{\alpha}, \overset{\circ}{\alpha}') > 0$ such that $[e', z] \neq 0$, which is possible as $\overset{\circ}{\mathfrak{g}}$ is simple. At this point, if $z \in \overset{\circ}{\mathfrak{h}}$, we fix a basis $h_1, h_2, \dots$ of $\overset{\circ}{\mathfrak{h}}$ such that $[e', h_i] = 0$ for $i \geq 2$. Since $\{\tau(\gamma) \mid \gamma \in \Gamma\}$ is dense in $\mathbb{R}$, there exists $\gamma' \in \Gamma$ such that $\tau(\operatorname{rt}(x)) < -(\tau(\overset{\circ}{\alpha}') + \tau(\gamma')) < 0$ for any $x \in B$ entering a monomial $u_i$. It is not difficult to check that $(e' \otimes t^{\gamma' + K\gamma}) \cdot X = \sum_j u_j' v_j'$, where $v_j'$ are among $v_i$ and the height of each of the elements $u_j'$ is smaller than the height of $u_1$. This proves the existence of a vector in $N$ as required. We may furthermore, assume that all monomials $u_i$ are of the same degree.

Indeed, consider $h$ and $f \in \mathring{\mathfrak{g}}$ so that $\{e, h, f\}$ is a canonical $sl(2)$–basis. Then, repeating the procedure above, we can find a nonzero vector $X' = (\sum_i u_i' v_i') \cdot 1_\lambda$ for which $u_i'$ are monomials in $\{f \otimes t^{\gamma'} \in \mathfrak{n}^- \,|\, \gamma' \in \Gamma\}$. Now the monomials $u_i'$ are necessarily of the same degree.

Assume that $X = (\sum_i u_i v_i) \cdot 1_\lambda \in N$, where $u_i$ are (distinct) monomials of the same degree $d$ in $\{e \otimes t^{\gamma'} \in \mathfrak{n}^- \,|\, \gamma' \in \Gamma\}$. As in Case 1 we need to prove that $N_\mathfrak{k} \neq 0$, and, if $\lambda(\Gamma_0) \neq 0$, then $v_i \in N_\mathfrak{k}$. The proof will be carried by induction on the maximal degree of a monomial $u_i$. Set $\Gamma_X := \{\gamma' \in \Gamma \,|\, e \otimes t^{\gamma'}$ occurs in a monomial $u_i\}$. Let $\gamma_0 \in \Gamma_X$ be the element with the following properties: $\tau(\gamma_0) = \min_{\gamma' \in \Gamma} \tau(\gamma')$ and if $\tau(\gamma_0) = \tau(\gamma')$ for some $\gamma' \in \Gamma_X$, then $\gamma_i - \gamma_0 \in \Delta^+$. This simply means that, for any integer $K$ and any $\gamma' \in \Gamma_X$, $[f \otimes t^{-\gamma_0 + K\gamma}, e \otimes t^{\gamma'}]$ is either in $\mathfrak{n}^+$ or in $\mathfrak{k} \cap \mathfrak{b}^-$. Assume furthermore, that $u_1 = x_1 \dots x_l$, where exactly $p$ of the vectors $x_1, \dots, x_l$ equal $e \otimes t^{\gamma_0}$, and $x_1$ is one of them. Consider $X' := (f \otimes t^{-\gamma_0 + K\gamma}) \cdot X$. The reader will verify easily that if $X' = (\sum_i u_i' v_i') \cdot 1_\lambda$, then, after a possible relabeling, $u_1' = x_2 \dots x_l$, $v_1' = p(h \otimes t^{K\gamma}) v_1$ and all $u_i'$ are polynomials in $\{e \otimes t^{\gamma'} \in \mathfrak{n}^- \,|\, \gamma' \in \Gamma\}$ of degree $d - 1$. An obvious induction argument shows that $N_\mathfrak{k} \neq 0$. In order to prove that $v_1 \in N_\mathfrak{k}$ we proceed as in the last paragraph of the proof of Case 1, Subcase 1. □

**Example 5.** Let $\mathring{\mathfrak{g}}$ be simple, $\Gamma_0 = \Gamma$ and $\mathring{\mathfrak{k}} = \mathring{\mathfrak{h}}$. In this case Theorem 1 has been proved in [FK]. The other previously treated cases are:

- $\operatorname{rk} \Gamma = 1$, arbitrary $\mathring{\mathfrak{g}}$ and arbitrary $\mathring{\mathfrak{k}}$, [CFM],
- $\operatorname{rk} \Gamma = 2$, $\operatorname{rk} \Gamma_0 = 1$, arbitrary $\mathring{\mathfrak{g}}$ and $\mathring{\mathfrak{k}} = \mathring{\mathfrak{g}}$, [CF],
- $\operatorname{rk} \Gamma = 2$, $\operatorname{rk} \Gamma_0 = 1$, $\mathring{\mathfrak{g}} = sl(2)$ and arbitrary $\mathring{\mathfrak{k}}$ (i.e., $\mathring{\mathfrak{k}} = \mathring{\mathfrak{g}}$ or $\mathring{\mathfrak{k}} = \mathring{\mathfrak{h}}$), [FK].

**Example 4 revisited.** In the notations of Example 4, $\Gamma_0 = \mathbb{Z}\delta_1$ is the lattice corresponding both to $\mathfrak{k}$ and $\mathfrak{k}_2$. Starting with the first Borel subalgebra we can apply Theorem 1 under the condition that $\lambda(\delta_1) \neq 0$. This reduces the study of subquotients of $M_\mathfrak{g}(\lambda)$ to the study of subquotients of Verma modules of $\mathfrak{k} \cong sl(2)[1]$ for a standard Borel subalgebra, which are known to have a rich and interesting structure. For the second Borel subalgebra we can apply the reduction theorem twice under the same condition. (Note that in general one needs a stronger condition to apply a second reduction.) This shows that the structure of the subquotients of $M_\mathfrak{g}(\lambda)$ is much simpler in this case, as all Verma modules of the Heisenberg algebra $\mathfrak{k}_2$ with nonzero highest weight are irreducible.

# 5 Equivalence of categories

In this final section we introduce an analog of the category $\mathcal{O}$ for toroidal Lie algebras and extend Theorems 1 and 2 to appropriate equivalences of categories.

Consider first the case when $\overset{\circ}{\mathfrak{g}}$ is semisimple. Let $\mathcal{O}_{\mathfrak{g}}$ denote the full subcategory of the category $\mathfrak{g}$–mod whose objects $M$ admit an accending $\mathfrak{g}$–module filtration $\{M_i\}$ such that the quotients $M_i/M_{i+1}$ are isomorphic to subquotients of Verma modules. The category $\mathcal{O}_{\mathfrak{k}}$ is defined analogously. Since $\mathfrak{z}$ acts via a character on each subquotient of a highest weight module, $\mathfrak{z}$ acts locally finitely on any object of $\mathcal{O}_{\mathfrak{g}}$ or $\mathcal{O}_{\mathfrak{k}}$. Both categories $\mathcal{O}_{\mathfrak{g}}$ and $\mathcal{O}_{\mathfrak{k}}$ are closed under countable direct sums.

Let $\mathcal{O}_{\mathfrak{g}}(\nu)$ and $\mathcal{O}_{\mathfrak{k}}(\nu)$ be the full subcategories of $\mathcal{O}_{\mathfrak{g}}$ and $\mathcal{O}_{\mathfrak{k}}$ respectively, such that their objects have the property that $\mathfrak{z}$ acts via the homomorphism $\nu : \mathfrak{z} \to \mathbb{C}$ on all quotients of their defining filtrations. The category $\mathcal{O}_{\mathfrak{g}}$ is then the full subcategory of the direct product of categories $\times_{\nu \in \mathfrak{z}^*} \mathcal{O}_{\mathfrak{g}}(\nu)$ whose objects are countably dimensional vector spaces over $\mathbb{C}$. A similar statement holds for $\mathcal{O}_{\mathfrak{k}}$ as well.

Induction provides a functor

$$I : \mathcal{O}_{\mathfrak{k}} \to \mathcal{O}_{\mathfrak{g}}, \quad N \mapsto U(\mathfrak{g}) \otimes_{U(\mathfrak{k} \oplus \mathfrak{n}^+)} N,$$

where $\mathfrak{n}^+$ acts trivially on $N$. The canonical image of $N$ in $U(\mathfrak{g}) \otimes_{U(\mathfrak{k} \oplus \mathfrak{n}^+)} N$ consists of $\mathfrak{n}^+$–invariants and, hence, $I(N)$ is generated by its $\mathfrak{n}^+$–invariants. Clearly, the restriction of $I$ on $\mathcal{O}_{\mathfrak{k}}(\nu)$, which we also denote by $I$, is a well defined functor $I : \mathcal{O}_{\mathfrak{k}}(\nu) \to \mathcal{O}_{\mathfrak{g}}(\nu)$.

For $\nu(\Gamma_0) \neq 0$ we will explicitly construct a functor adjoint to $I$. We start with the following lemma.

**Lemma 1** *Let $M$ be a subquotient of $M_{\mathfrak{g}}(\lambda)$. If $\lambda(\Gamma_0) \neq 0$, then $M^{\mathfrak{n}^+} = M_{\mathfrak{k}}$, where $M^{\mathfrak{n}^+}$ are the $\mathfrak{n}^+$–invariants of $M$ and $M_{\mathfrak{k}}$ is as in Section 4.*

**Proof.** It suffices to prove the lemma for quotients of $M_{\mathfrak{g}}(\lambda)$. Let $M = M_{\mathfrak{g}}(\lambda)/K$. By definition, $(M_{\mathfrak{g}}(\lambda)/K)_{\mathfrak{k}} = M_{\mathfrak{g}}(\lambda)_{\mathfrak{k}}/K_{\mathfrak{k}}$ and, by Theorem 1, $M_{\mathfrak{g}}(\lambda)^{\mathfrak{n}^+} = M_{\mathfrak{g}}(\lambda)_{\mathfrak{k}}$ and $K^{\mathfrak{n}^+} = K_{\mathfrak{k}}$. Hence, it is enough to show that $(M_{\mathfrak{g}}(\lambda)/K)^{\mathfrak{n}^+} = M_{\mathfrak{g}}(\lambda)^{\mathfrak{n}^+}/K^{\mathfrak{n}^+}$.

The inclusion $(M_{\mathfrak{g}}(\lambda)/K)^{\mathfrak{n}^+} \supset M_{\mathfrak{g}}(\lambda)^{\mathfrak{n}^+}/K^{\mathfrak{n}^+}$ is obvious. The opposite inclusion follows from Theorem 1. Indeed, assume to the contrary, that $0 \neq m \in (M_{\mathfrak{g}}(\lambda)/K)^{\mathfrak{n}^+} \notin M_{\mathfrak{g}}(\lambda)^{\mathfrak{n}^+}/K^{\mathfrak{n}^+}$. Then $m$ generates a proper submodule $M'$ of $M$ with the property that $M'_{\mathfrak{k}} \cap M_{\mathfrak{k}} = 0$. The preimage $M''$ of $M'$ in $M_{\mathfrak{g}}(\lambda)$ has the property that $M''_{\mathfrak{k}} = K_{\mathfrak{k}}$, which is in contradiction with Theorem 1 (ii) as $M'' \neq K$. □

Lemma 1 implies now that for $\nu(\Gamma_0) \neq 0$ we have a well defined functor

$$R : \mathcal{O}_{\mathfrak{g}}(\nu) \to \mathcal{O}_{\mathfrak{k}}(\nu), \quad M \mapsto M^{\mathfrak{n}^+}.$$

Indeed, in order to see that $M^{\mathfrak{n}^+}$ is an object of $\mathcal{O}_{\mathfrak{k}}(\nu)$ we consider a filtration of $M$ whose associated quotients are subquotients of Verma $\mathfrak{g}$–modules. The consequtive application of Lemma 1 to the quotients $M_i/M_{i-1}$ ensures that the associated quotients of the filtration $\{M_i^{\mathfrak{n}^+}\}$ of $M^{\mathfrak{n}^+}$ are subquotients of Verma $\mathfrak{k}$–modules.

If $\nu(\Gamma_0) = 0$, it is not true in general that, for any object $M$ of $\mathcal{O}_{\mathfrak{g}}(\nu)$, $R(M) = M^{\mathfrak{n}^+}$ is an object of $\mathcal{O}_{\mathfrak{k}}(\nu)$. Here is an example.

**Example 6.** Let $\mathfrak{g} = sl(2)[1]$, let $\mathfrak{b}$ be an imaginary Borel subalgebra, i.e. $\mathfrak{k}$ is the Heisenberg algebra, and let $\lambda = 0$. The module $M_{\mathfrak{g}}(0)$ is well understood, see [F2]. Denote by $N$ the maximal proper submodule of $M_{\mathfrak{k}}(0)$, let $K$ be the $\mathfrak{g}$–submodule of $M_{\mathfrak{g}}(0)$ generated by $N$ and set $M := M_{\mathfrak{g}}(0)/K$. A direct computation shows that $M^{\mathfrak{n}^+}$ contains a $\mathfrak{k}$–submodule isomorphic to $\mathfrak{n}^-$. Therefore, if $M^{\mathfrak{n}^+}$ admitted a filtration $\{(M^{\mathfrak{n}^+})_i\}$ as required, $\mathfrak{n}^-$ would have nontrivial $\mathfrak{k} \cap \mathfrak{b}$–invariants. However, the $\mathfrak{k}$–module $\mathfrak{n}^-$ has no $\mathfrak{k} \cap \mathfrak{b}$–invariants, and hence $M^{\mathfrak{n}^+}$ is not an object of the category $\mathcal{O}_{\mathfrak{k}}(0)$.

**Theorem 3** *Let $\overset{\circ}{\mathfrak{g}}$ be semisimple and $\nu(\Gamma_0) \neq 0$.*

(i) *The functor $I : \mathcal{O}_{\mathfrak{k}}(\nu) \to \mathcal{O}_{\mathfrak{g}}(\nu)$ is a left adjoint to the functor $R : \mathcal{O}_{\mathfrak{g}}(\nu) \to \mathcal{O}_{\mathfrak{k}}(\nu)$, i.e. $R \circ I$ is naturally isomorphic to the identity functor on $\mathcal{O}_{\mathfrak{k}}(\nu)$.*

(ii) *The functors $R$ and $I$ are mutually inverse equivalences of $\mathcal{O}_{\mathfrak{k}}(\nu)$ and the full subcategory $\mathcal{O}^{\mathfrak{k}}_{\mathfrak{g}}(\nu)$ of $\mathcal{O}_{\mathfrak{g}}(\nu)$ whose objects are generated as $\mathfrak{g}$–modules by their $\mathfrak{n}^+$–invariants.*

**Proof.** (i) Let $N$ be an object of $\mathcal{O}_{\mathfrak{k}}(\nu)$ with a filtration $\{N_i\}$. Clearly, there is a canonical injection of $\mathfrak{k}$–modules

$$\Phi : N \to (U(\mathfrak{g}) \otimes_{U(\mathfrak{k} \oplus \mathfrak{n}^+)} N)^{\mathfrak{n}^+}.$$

To see that $\Phi$ is a surjection, assume the contrary. Then for some minimal $i$ there is a nonzero vector $m \in M_i^{\mathfrak{n}^+} := (U(\mathfrak{g}) \otimes_{U(\mathfrak{k} \oplus \mathfrak{n}^+)} N_i)^{\mathfrak{n}^+}$, $m \notin \Phi(N_i)$. Hence the projection of $m$ in $M_i/M_{i-1}$ is a nontrivial $\mathfrak{n}^+$–invariant vector such that $m \notin (M_i/M_{i-1})_{\mathfrak{k}}$. This contradicts Lemma 1, therefore $\Phi$ is an isomorphism.

The above argument shows that $R \circ I$ is naturally isomorphic to the identity functor on the objects of $\mathcal{O}_{\mathfrak{k}}(\nu)$. A straightforward checking shows that this is true on morphisms too.

To show (ii), notice that for any object $M$ of $\mathcal{O}_{\mathfrak{g}}(\nu)$ there is a canonical morphism of $\mathfrak{g}$–modules

$$\Psi : U(\mathfrak{g}) \otimes_{U(\mathfrak{k} \oplus \mathfrak{n}^+)} M^{\mathfrak{n}^+} \to M.$$

We claim that $\Psi$ is injective. To see this it suffices to prove that $\Psi$ is injective on $U(\mathfrak{g}) \otimes_{U(\mathfrak{k} \oplus \mathfrak{n}^+)} M_i^{\mathfrak{n}^+}$ for every $i$, $\{M_i\}$ being the defining filtration on $M$. The latter is established by a simple inductive argument based on Theorem 1. Finally, in order to assert that $\Psi$ is surjective one has to assume that $M$ is an object of $\mathcal{O}^{\mathfrak{k}}_{\mathfrak{g}}(\nu)$. This implies (ii) modulo some details that the reader will easily fill in. □

Here is an explicit example of an object in $\mathcal{O}_{\mathfrak{g}}(\nu)$ which is not an object of $\mathcal{O}^{\mathfrak{k}}_{\mathfrak{g}}(\nu)$.

**Example 7.** Let $\mathfrak{g} = sl(2)[1]$, cf. Example 1. Set $e(i) := e \otimes t^i$, $f(i) := f \otimes t^i$, $h(i) = h \otimes t^i$, for $i \in \mathbb{Z}$. Here $\{e, f, h\}$ is a standard basis of $sl(2)$ and $\kappa$ is normalized so that $\kappa(h, h) = 2$. Then the explicit commutation relations of $sl(2)[1]$ are as follows: $[e(i), f(j)] = h(i + j) + i\delta_{i,-j}c$, $[h(i), e(j)] = 2e(i + j)$, $[h(i), f(j)] = -2f(i + j)$, $[h(i), h(j)] = 2i\delta_{i,-j}c$,

$[e(i),e(j)] = [f(i),f(j)] = 0$, $[d,e(i)] = ie(i)$, $[d,h(i)] = ih(i)$, $[d,f(i)] = if(i)$, $[c,e(i)] = [c,h(i)] = [c,f(i)] = [c,d] = 0$. Fix the Borel subalgebra $\mathfrak{b} = \mathfrak{h} \oplus_{i>0} \mathbb{C}h(i) \oplus_i \mathbb{C}e(i)$. For the corresponding triangular decomposition $\mathfrak{g} = \mathfrak{n}^- \oplus \mathfrak{k} \oplus \mathfrak{n}^+$ we have $\mathfrak{n}^- = \oplus_i \mathbb{C}f(i)$, $\mathfrak{k} = \oplus_i \mathbb{C}h(i) \oplus \mathbb{C}c \oplus \mathbb{C}d$ and $\mathfrak{n}^+ = \oplus_i \mathbb{C}e(i)$.

First we construct an indecomposable $\mathfrak{k} \oplus \mathfrak{n}^+$–module $M_0$ of length 2. Let $\mathbb{C}[s_1, s_2, ...]$ be the ring of polynomials in infinitely many variables $\{s_i\}_{i=1,2,...}$, and let $W = \mathbb{C}v \oplus \mathbb{C}u$ be a two dimensional vector space. Set

$$h(i) \cdot p(s) := \begin{cases} 2i\frac{\partial}{\partial s_i}p(s) & \text{for} \quad i < 0, \\ s_i p(s) & \text{for} \quad i > 0, \end{cases}$$

$c \cdot p(s) := p(s)$, and $d \cdot p(s) := (\sum_{i=1}^{\infty} i\frac{\partial}{\partial s_i})p(s)$ for any $p(s) \in \mathbb{C}[s_1, s_2, \dots]$. Then $M' := \mathbb{C}[s_1, s_2, \dots] \otimes W$ is a $\oplus_{i\neq 0}\mathbb{C}h(i) \oplus \mathbb{C}c \oplus \mathbb{C}d$–module with the above action on $\mathbb{C}[s_1, s_2, \dots]$ and with trivial action on $W$. By putting $h(0) \cdot (p(s) \otimes v) := 0$ and $h(0) \cdot (p(s) \otimes u) := -2p(s) \otimes u$, we define a $\mathfrak{k}$–module structure on $M'$. Clearly, $M'$ decomposes into a direct sum of the irreducible $\mathfrak{k}$–modules $\mathbb{C}[s_1, s_2, \dots] \otimes v$ and $\mathbb{C}[s_1, s_2, \dots] \otimes u$.

Below we will extend the $\mathfrak{k}$–module structure on $M'$ to a $\mathfrak{k} \oplus \mathfrak{n}^+$–module structure such that $(M')^{\mathfrak{n}^+} = \mathbb{C}[s_1, s_2, \dots] \otimes v$. Then $M := I(M) = U(\mathfrak{g}) \otimes_{U(\mathfrak{k}\oplus\mathfrak{n}^+)} M'$ will be an indecomposable $\mathfrak{g}$–module with composition series $0 \subset M_1 \subset M$ with $M/M_1 \cong M_2$, where $M_1 := I(\mathbb{C}[s_1, s_2, \dots] \otimes v)$ and $M_2 := I(\mathbb{C}[s_1, s_2, \dots] \otimes u)$, $\mathfrak{n}^+$ acting trivially on $\mathbb{C}[s_1, s_2, \dots] \otimes v$ and $\mathbb{C}[s_1, s_2, \dots] \otimes u$. Indeed, $\mathbb{C}[s_1, s_2, \dots] \otimes v$ and $\mathbb{C}[s_1, s_2, \dots] \otimes u$ are irreducible $\mathfrak{k}$–modules on which $c$ acts nontrivially, and hence, by Theorem 1, both $M_1$ and $M_2$ are irreducible $\mathfrak{g}$–modules. Finally, $M^{\mathfrak{n}^+} = \mathbb{C}[s_1, s_2, \dots] \otimes v$, and hence $(I \circ R)(M) \cong M_1$.

To construct the $\mathfrak{k} \oplus \mathfrak{n}^+$–structure on $M'$ we first set $e(n) \cdot (p(s) \otimes v) := 0$ and recall that Schur's polinomials $S_n$ are given by their generating function $\exp(\sum_{i\geq 1} s_i x^i) = \sum_{n\in\mathbb{Z}} S_n(s_1, s_2, \dots)x^n$. (Note that $S_n = 0$ for $n < 0$.) From the definition of $S_n$ we obtain

$$\frac{\partial}{\partial s_i} S_n(s) = S_{n-i}(s).$$

We show now that there exists a unique $\mathfrak{k} \oplus \mathfrak{n}^+$–module structure on $M'$ which extends its $\mathfrak{k}$–module structure and such that $e(0) \cdot (1 \otimes u) = 1 \otimes v$. The uniqueness is clear. For the existence, we first use the commutation relations $[h(i), e(-n)] = 2e(i-n)$ with $i > 0$ and $n > 0$ to derive the equality $e(-n) \cdot (1 \otimes u) = S_n(\bar{s}) \otimes v$, where $\bar{s}$ stands for $s_1, s_2/2, s_3/3, \dots$. Furthermore, we extend the latter formula to

$$e(-n) \cdot (s_{i_1} s_{i_2} \dots s_{i_l} \otimes u) = \left( \sum_{L \subset K} (-2)^{\#L} \left( \prod_{i \in K \setminus L} s_i \right) S_{n - \sum_{i\in L} i}(\bar{s}) \right) \otimes v, \tag{7}$$

where $n \in \mathbb{Z}$, $K$ is the set of all indices $i_1, i_2, \dots, i_l$ with possible repetitions, and $\#L$ stands for the number of elements of $L$. We leave it to the reader to check that (7) together with the equality $e(n) \cdot (p(s) \otimes v) = 0$ determines a $\mathfrak{k} \oplus \mathfrak{n}^+$–module structure on the $\mathfrak{k}$–module

$M'$. Finally, since $M'$ has length 2 as a $\mathfrak{k}$–module, $(M')^{\mathfrak{n}^+}$ is a $\mathfrak{k}$–submodule of $M'$, and $e(0)\cdot(1\otimes u)=1\otimes v$, we conclude that $(M')^{\mathfrak{n}^+}=\mathbb{C}[s_1,s_2,\dots]\otimes v$.

Recall that the Borel subalgebra $\mathfrak{b}$ of $\mathfrak{g}$ determines a strictly decreasing sequence of toroidal Lie algebras (3). In order to be able to construct a chain of adjoint functors between $\mathfrak{k}_i$–modules and $\mathfrak{k}_{i+1}$–modules, we need a generalization of Theorem 3 in the case of a reductive $\overset{\circ}{\mathfrak{g}}$. Theorem 2 enables us to modify the category $\mathcal{O}_{\mathfrak{g}}(\nu)$ and the functor $I$ accordingly.

Assume now that $\overset{\circ}{\mathfrak{g}}$ is reductive. Let $\widetilde{\mathcal{O}_{\mathfrak{g}}}(\nu)$ be the full subcategory of $\mathcal{O}_{\mathfrak{g}}(\nu)$ on whose objects $Z^-(\nu)$ acts trivially, and let $\tilde{I}$ be the functor

$$\tilde{I}:\mathcal{O}_{\mathfrak{k}}(\nu)\to\widetilde{\mathcal{O}_{\mathfrak{g}}}(\nu),\quad \tilde{N}\mapsto U(\tilde{\mathfrak{g}})\otimes_{U(\mathfrak{k}\oplus\mathfrak{n}^+)}N,$$

where $\mathfrak{n}^+$ acts trivially on $N$. Using Theorem 2 one establishes that for any object $M$ of $\widetilde{\mathcal{O}_{\mathfrak{g}}}(\nu)$, $R(M)$ is an object of $\mathcal{O}_{\mathfrak{k}}(\nu)$. The main result for reductive $\overset{\circ}{\mathfrak{g}}$ is the following theorem.

**Theorem 4** *Let $\overset{\circ}{\mathfrak{g}}$ be reductive and $\nu(\Gamma_0)\neq 0$.*

(i) *The functor $\tilde{I}:\mathcal{O}_{\mathfrak{k}}(\nu)\to\widetilde{\mathcal{O}_{\mathfrak{g}}}(\nu)$ is a left adjoint to the functor $R:\widetilde{\mathcal{O}_{\mathfrak{g}}}(\nu)\to\mathcal{O}_{\mathfrak{k}}(\nu)$, i.e. $R\circ\tilde{I}$ is naturally isomorphic to the identity functor on $\mathcal{O}_{\mathfrak{k}}(\nu)$.*

(ii) *The functors $R$ and $\tilde{I}$ are mutually inverse equivalences of $\mathcal{O}_{\mathfrak{k}}(\nu)$ and the full subcategory $\widetilde{\mathcal{O}}^{\mathfrak{k}}_{\mathfrak{g}}(\nu)$ of $\widetilde{\mathcal{O}_{\mathfrak{g}}}(\nu)$ whose objects are generated as $\mathfrak{g}$–modules by their $\mathfrak{n}^+$–invariants.*

The proof of Theorem 4 is a straightforward modification of the proof of Theorem 3 based on Theorem 2.

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I.D.:
Department of Mathematics
Yale University
10 Hillhouse Ave.
New Haven, CT 06520
USA
dimitrov@math.yale.edu

V.F.:
Instituto de Matemática e Estatística
Universidade de São Paulo
Caixa Postal 66281
CEP 05315-970 São Paulo
Brasil
futorny@ime.usp.br

I.P.:
Department of Mathematics
University of California
Riverside, CA 92521

USA
penkov@math.ucr.edu